

Craniosynostosis Suture Patterns

Clinical Models for Brain Region Volumes

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Intro

- **Question: how are suture fusions related to brain volume abnormalities in 12 designated regions?**
- Explanatory model describes the typical relationship between suture fusions and regional volume deviations
 - Shows a distinct pattern for each suture fusion
- Predictive model to forecast future volume deviations based on suture type and previous measurements
 - Several models perform similarly for prediction

- We limit the analysis to infants less than 1 year old at the time of the picture.
 - Includes over 50% of the data
- We control for the demographics **Sex, Age at Image, Image Type (CT vs. 3D)**
 - In the Explanatory Model, we also control for the total head volume deviation

Explanatory Model

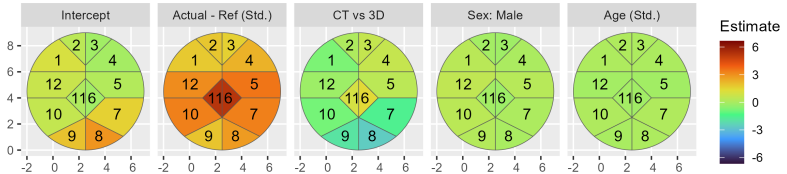
Suture Fusion Associations

$$\begin{aligned}(Y_{1i}, \dots, Y_{12i}) = & \sum_{k=1}^{12} \beta_{0,k} + \beta_{1,k}[\text{Metopic}]_i + \beta_{2,k}[\text{Sagittal}]_i + \\ & \beta_{3,k}[\text{Left Coronal}]_i + \beta_{4,k}[\text{Right Coronal}]_i + \\ & \beta_{5,k}[\text{Left Coronal} * \text{Right Coronal}]_i + \\ & \beta_{6,k}[\text{Total Head Volume}]_i + \beta_{7,k}[\text{Image Type}]_i + \\ & \beta_{8,k}[\text{Sex}]_i + \beta_{9,k}[\text{Age}]_i + \\ & (\epsilon_{1i}, \dots, \epsilon_{12i})\end{aligned}$$

- Explanatory model is a “multivariate” linear regression model:
 - 12 regions’ volume deviations are included as outcomes together
 - Captures covariance between suture fusion effects across regions
 - Allows for testing hypotheses about coefficients across regions (e.g. whether outcomes in regions 1-4 are related between the sagittal and metopic fusions)

Explanatory Control Coefficients

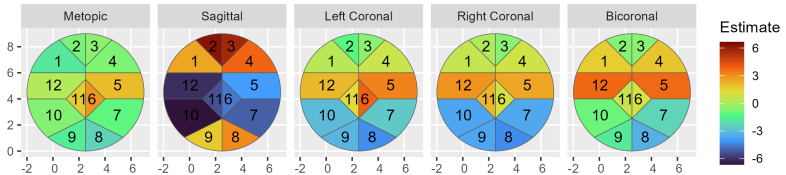
Expected Volume Deviations: Demographics



- No notable patterns associated with demographics

Explanatory Suture Coefficients

Expected Volume Deviations by Suture Type



- Metopic: larger volume in the middle regions
- Sagittal suture: larger in front + back / smaller on sides
- 1 coronal suture: smaller in back / larger in the opposite side middle + on the sides
- 2 coronal sutures: larger on sides + front

Explanatory Model Performance

Model Performance Statistics

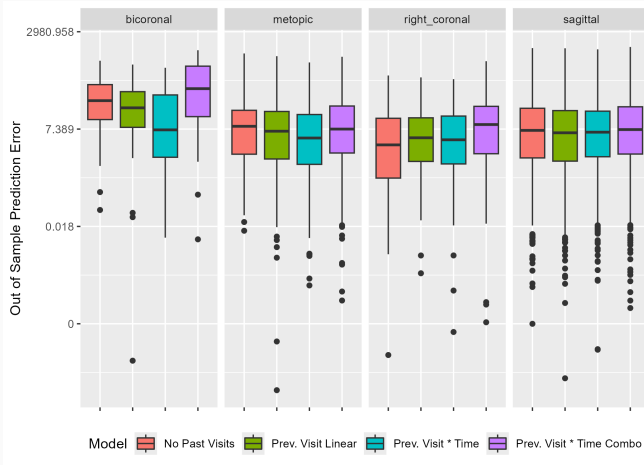
Region	Adj. R Squared	Adj. R Sq. w/o Sutures
1	0.497	0.37
2	0.443	0.24
3	0.436	0.225
4	0.558	0.417
5	0.578	0.404
6	0.784	0.636
7	0.353	0.239
8	0.525	0.387
9	0.411	0.308
10	0.369	0.205
11	0.782	0.624
12	0.609	0.311

Predictive Model

Prediction Model Description

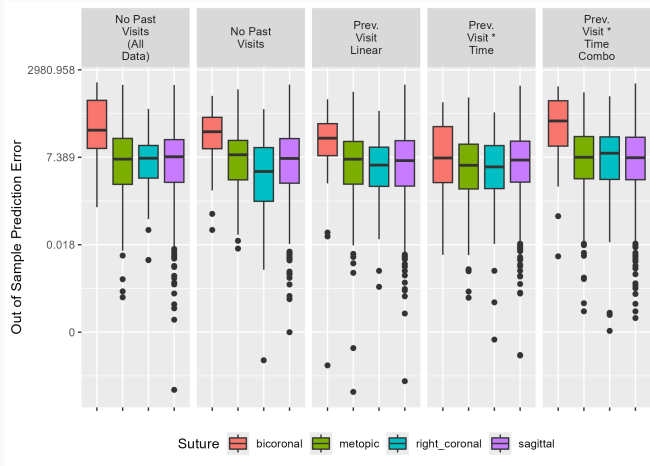
- Questions:
 - Can information collected at a patient's first visit be used to predict outcomes at their second?
 - Does this depend on the time between the first and second visit?
- We compare 4 predictive models:
 1. Region-specific Control (Sutures, Sex, Age at Image, and Image Type)
 - *A successful predictive model would perform better than this control*
 2. Region-specific Control + **Baseline Volume Deviation**
 3. **Region-specific Control** + Baseline Outcome *with Time Since Baseline*
 4. Joint model: Control + Baseline Outcome with Time Since Baseline *where baseline effect is estimated based on all regions together*
- We use leave-one-out cross validated predictions
 - For each patient with multiple visits, we exclude all of that patient's visits and train each model
 - We use these models and information from the patient's first visit to predict the outcome at the second visit

Prediction Performance by Suture



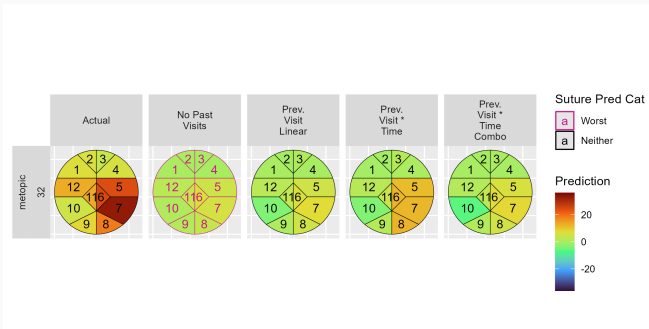
- No predictive model definitively has the smallest errors
- Region-specific model including time since baseline (teal) tends to perform as well as any

Prediction Performance by Model



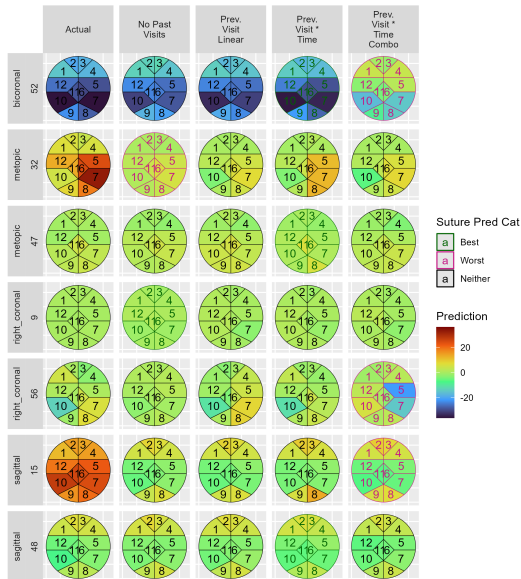
- Bicoronal patients tend to have the least accurate out-of-sample predictions, probably due to relatively few observations

Prediction Example



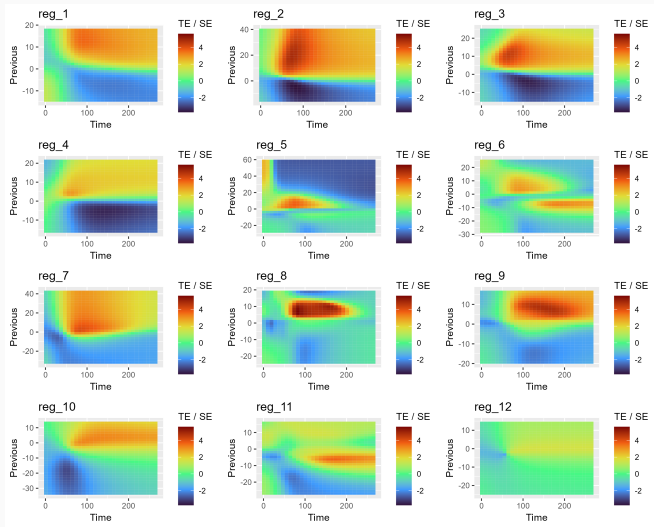
- We visualize these predictions on a specific patient
- Left column: actual volume deviations at the second visit
- Most models forecast this patient to follow the average metopic pattern
- The control model is the least accurate
- Models incorporating the patient's previous visit recognize the large volumes in regions 5-8

Additional Prediction Comparisons

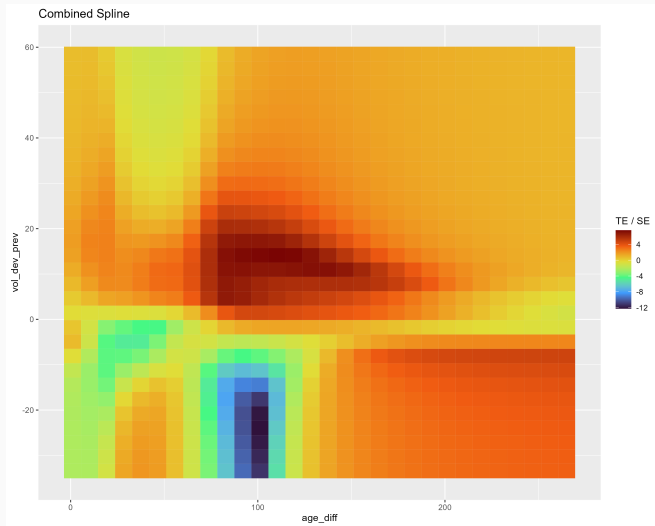


Appendix Slides

Previous Visit Effects i



Previous Visit Effects ii



Data Summaries

Suture Representation

Suture Type	Pictures	Patients	% of Patients	% of Included
no_suture	598	598	49.6%	49.6%
sagittal	371	371	30.8%	30.8%
metopic	106	106	8.8%	8.8%
bicoronal	55	55	4.6%	4.6%
right_coronal	45	45	3.7%	3.7%
left_coronal	31	31	2.6%	2.6%
Total	1206	1206	-	-

Sutures: Predictive Data

Suture Type	Pictures	Patients	% of Patients	% of Included
sagittal	65	55	62.5%	62.5%
metopic	25	20	22.7%	22.7%
right_coronal	11	7	8.0%	8.0%
bicoronal	5	5	5.7%	5.7%
left_coronal	1	1	1.1%	1.1%
Total	107	88	-	-